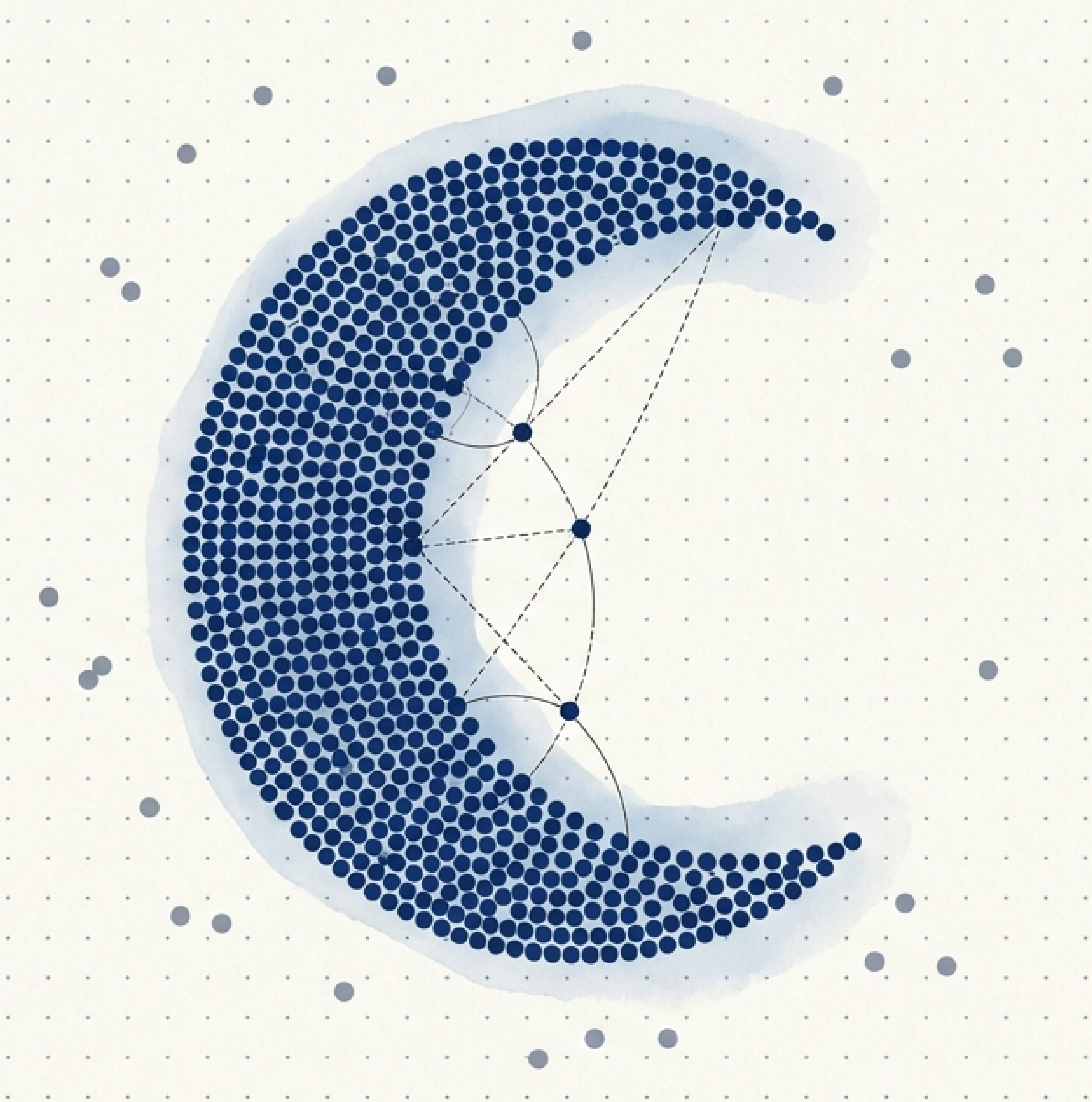
A Geometric Intuition for DBSCAN Clustering

Moving beyond centroids to map organic shapes and isolate anomalies in spatial data.



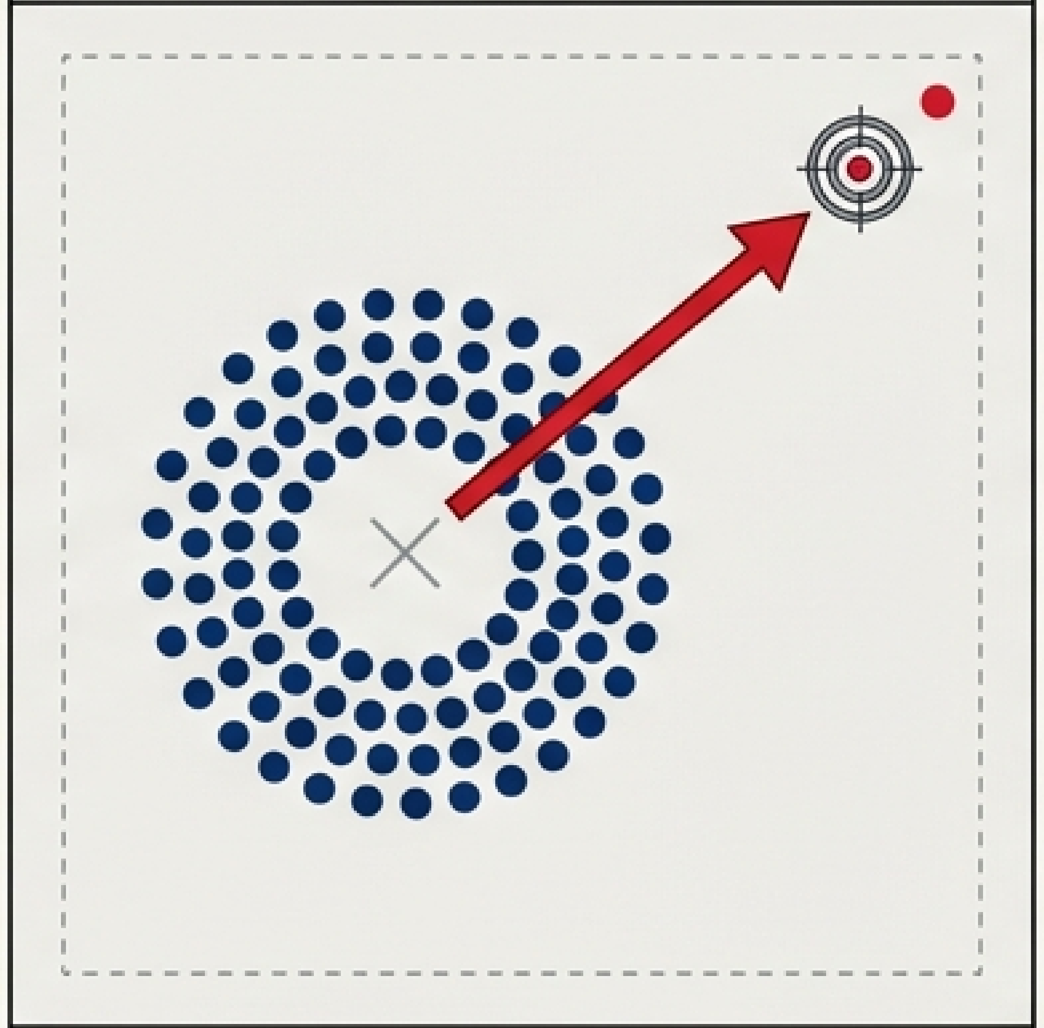
Traditional centroid models fail on complex topographies

The 'K' Dilemma

Ambiguous Elbow Curves

We must guess the number of clusters before the algorithm even starts.

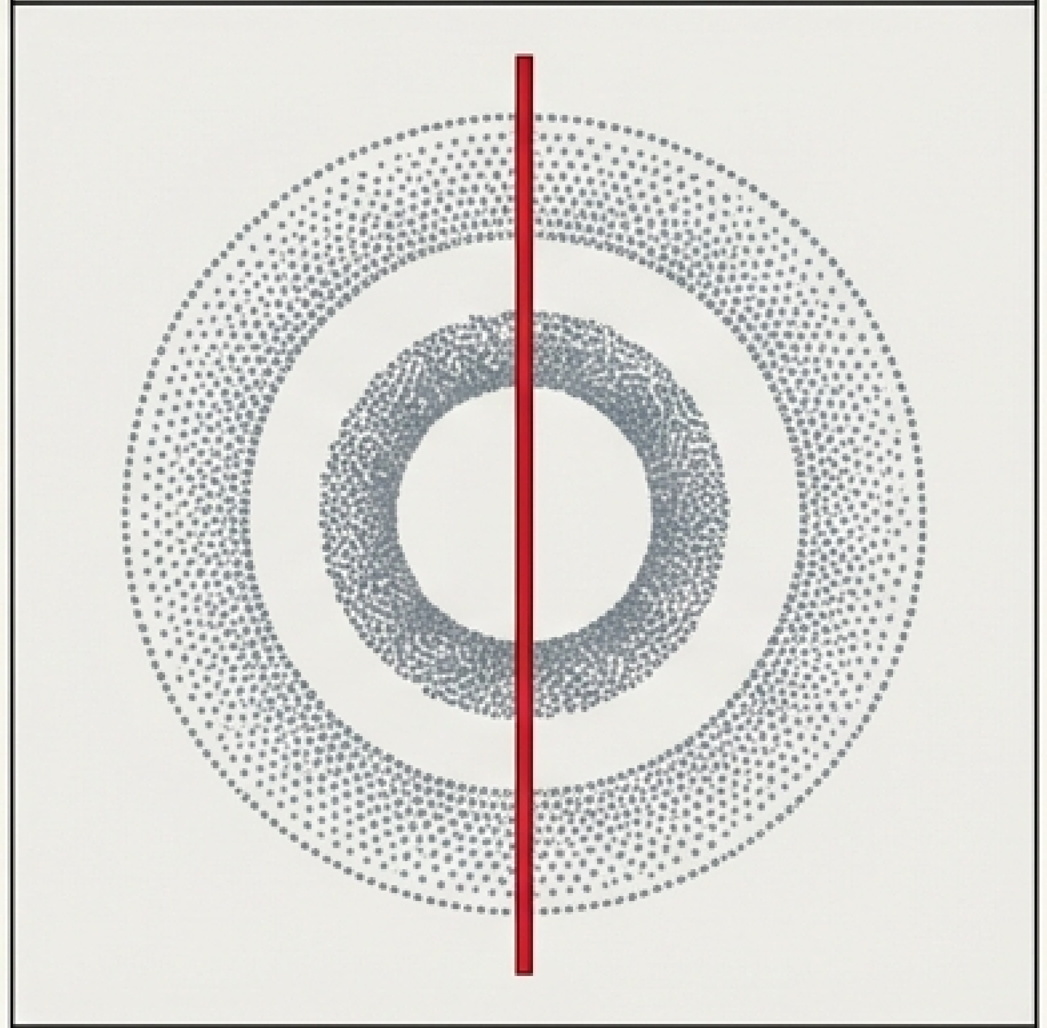
The Outlier Pull



High Outlier Sensitivity

A single anomaly distorts the entire cluster centre.

The Shape Trap



Blind to Arbitrary Shapes

Centroid models force spherical boundaries, breaking complex geometric patterns.

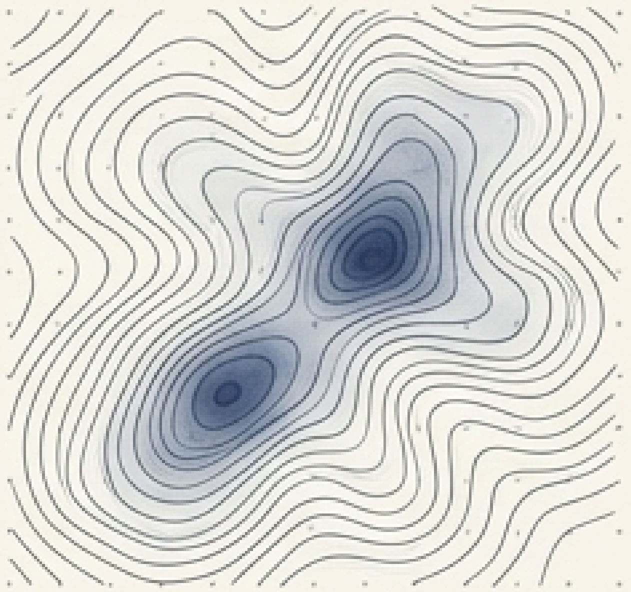
Moving from strict centroids to organic density

Centroid-Based (K-Means)

Core Logic:
Find the centre. Groups points by calculating their distance to a central, moving coordinate.

Ideal For:
Spherical, evenly distributed data.

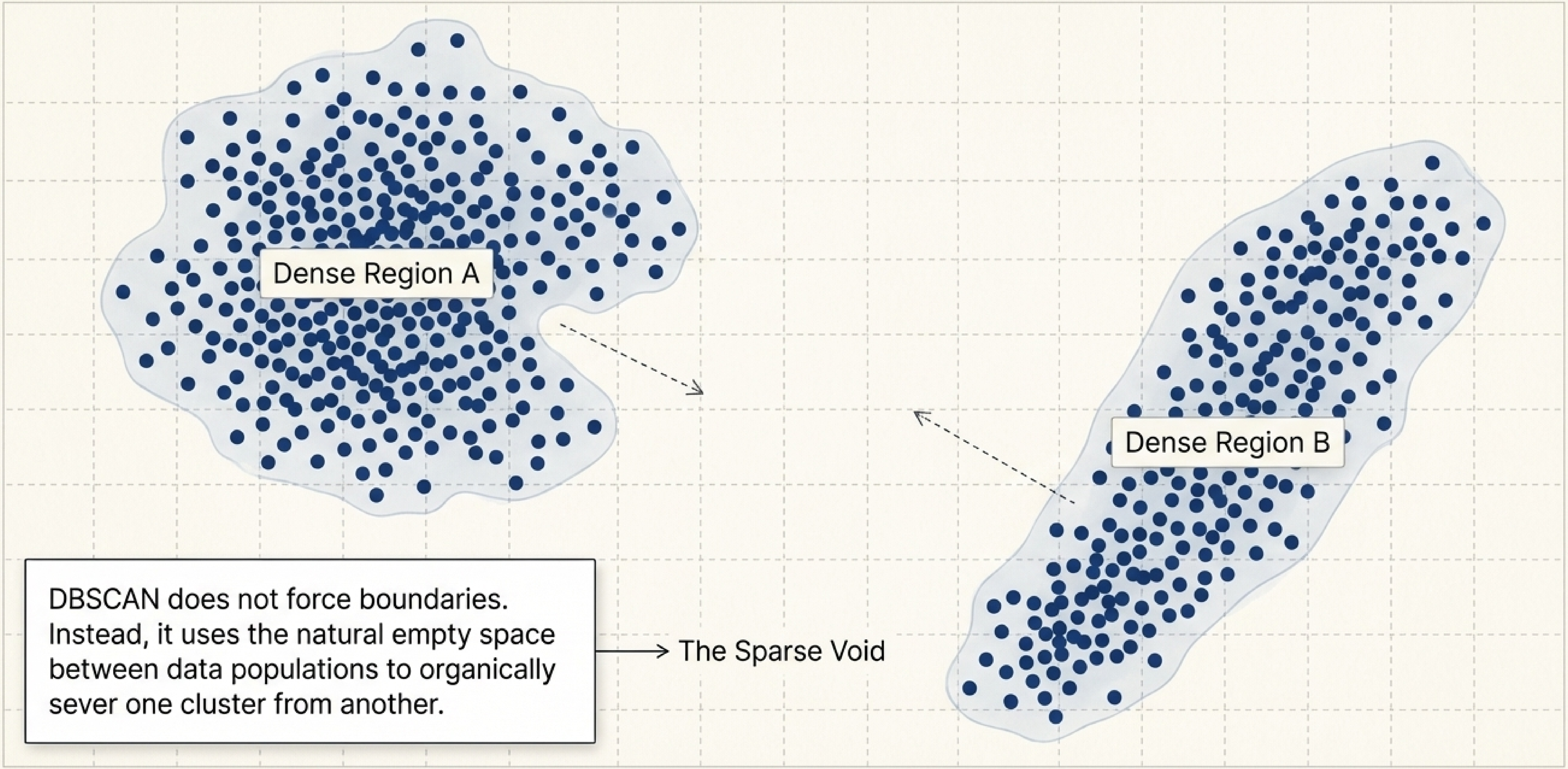
Density-Based (DBSCAN)



Core Logic:
Follow the crowd. Groups points based on local population density, separated by empty space.

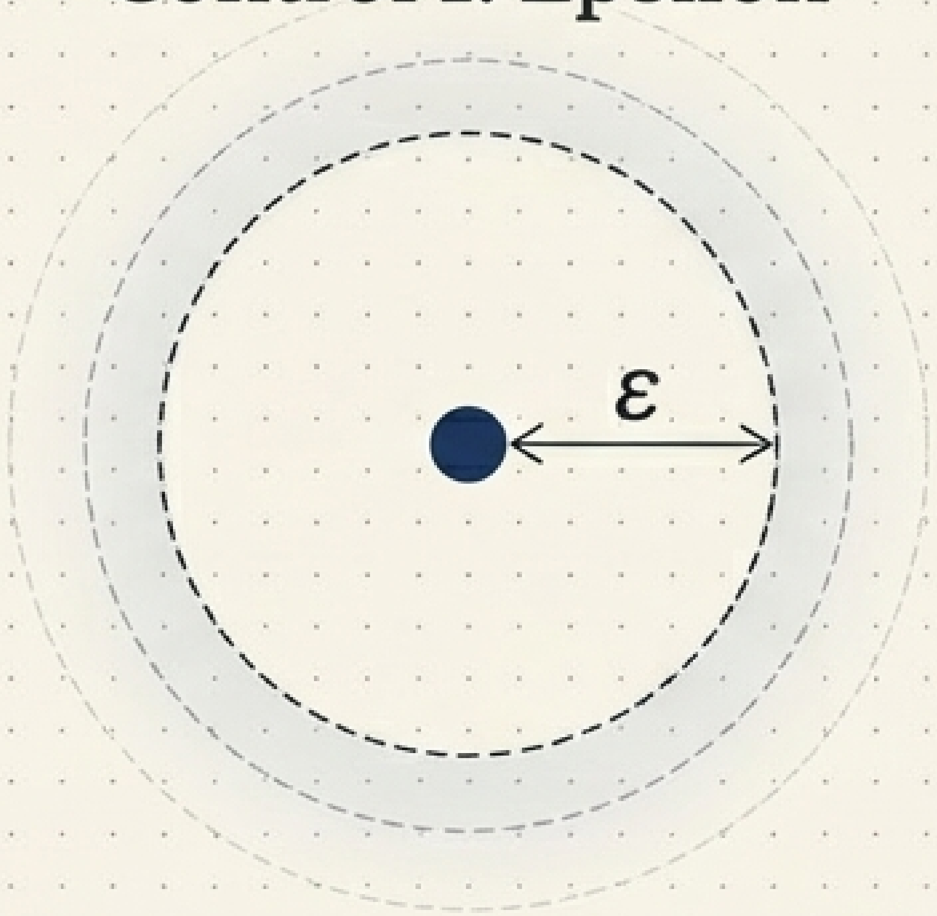
Ideal For:
Arbitrary shapes and datasets with heavy noise.

Sparse regions act as natural cluster boundaries



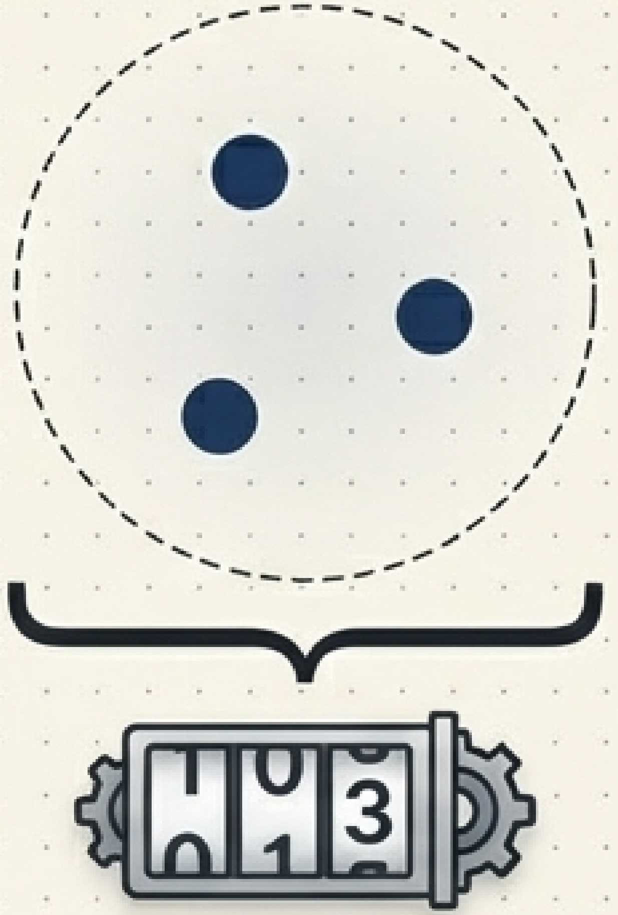
Two parameters define the rules of spatial density

Control 1: Epsilon



Epsilon radius
The maximum spatial distance of the neighbourhood surrounding a specific data point.

Control 2: MinSamples



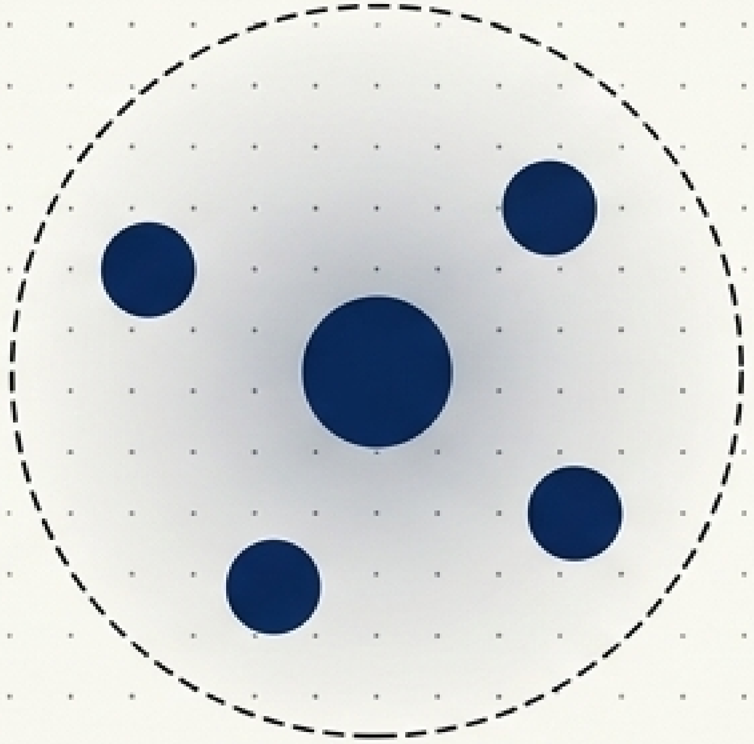
Minimum points threshold
The absolute minimum number of data points required within the Epsilon radius for that region to be officially classified as dense.

Every data point is assigned a permanent spatial identity

Assuming a threshold of MinSamples = 5

Cellular Anatomy

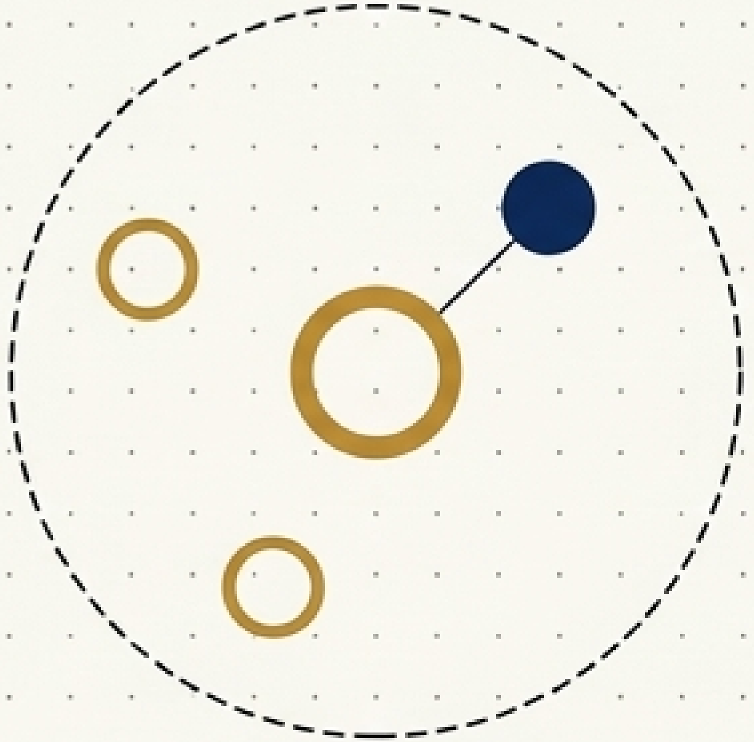
The Foundation



Core Point

Rule: Contains 5 or more points within its Epsilon neighbourhood. Forms the dense spine of a cluster.

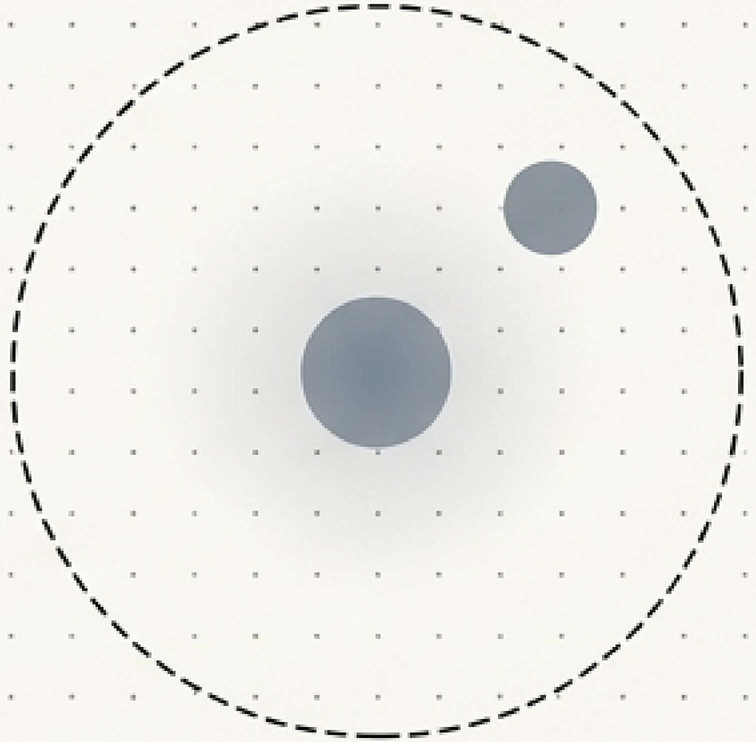
The Edge



Border Point

Rule: Contains fewer than 5 points, but its neighbourhood includes at least one Core Point. Forms the skin of a cluster.

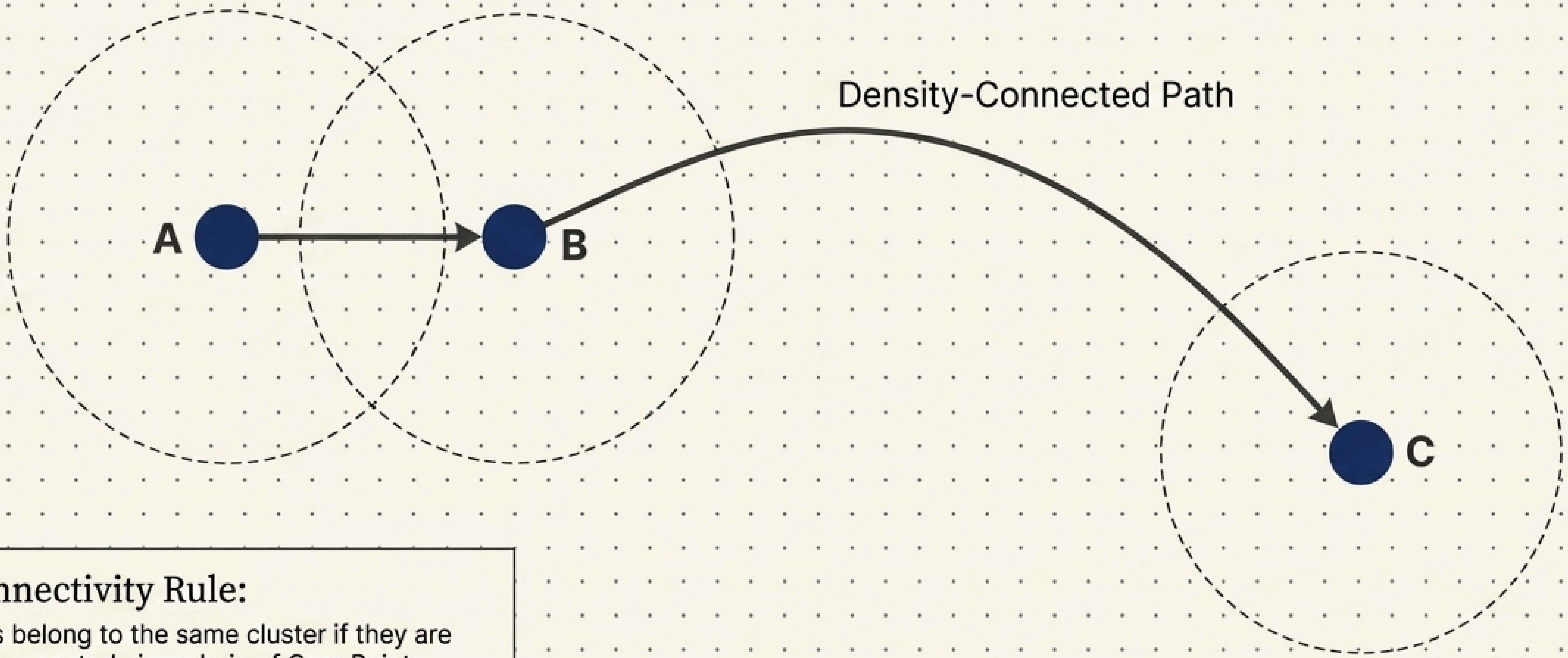
The Outlier



Noise Point

Rule: Contains fewer than 5 points and touches zero Core Points. Exists in the sparse void.

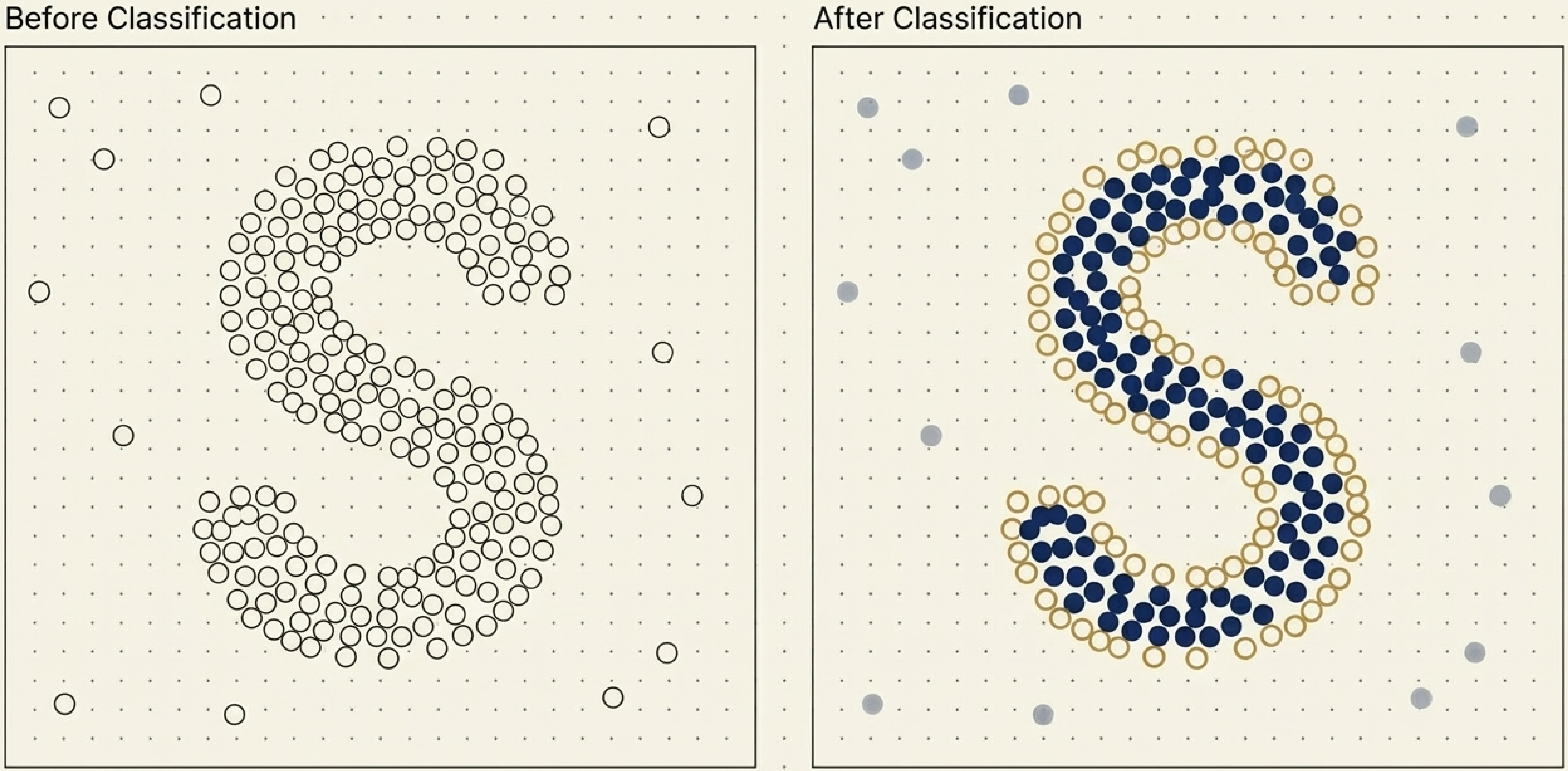
Overlapping neighbourhoods link micro-regions into macro-clusters



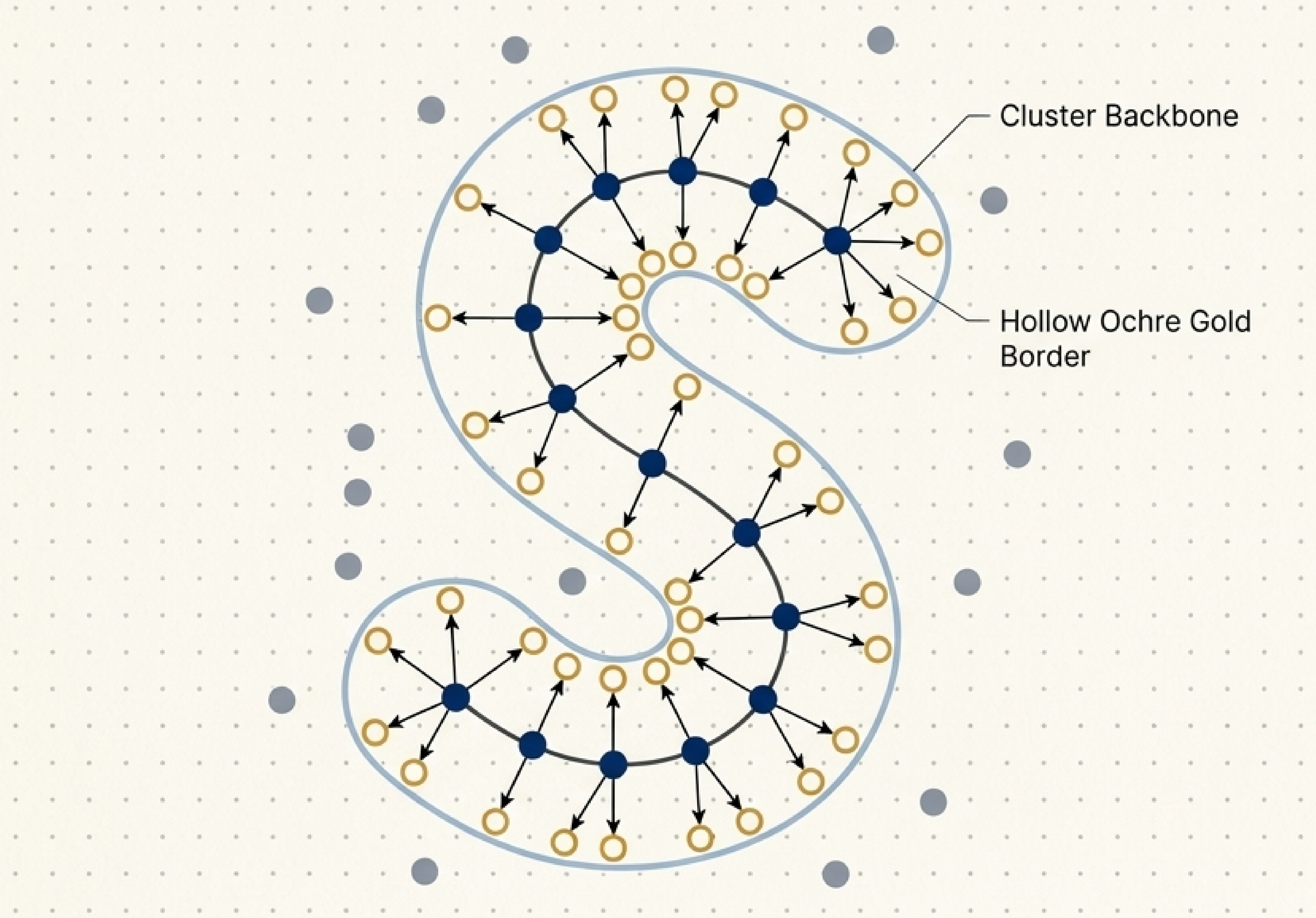
The Connectivity Rule:

Two points belong to the same cluster if they are indirectly connected via a chain of Core Points, where every adjacent pair is within Epsilon distance. If the distance exceeds the radius, or a point in the chain is not a Core Point, the path breaks.

Step 1: Scanning the landscape to classify every point



Steps 2 & 3: Chaining core networks and sweeping borders



The Formation Sequence

1. Unclustered Core points chain together via overlapping Epsilon radii to form the cluster skeleton.

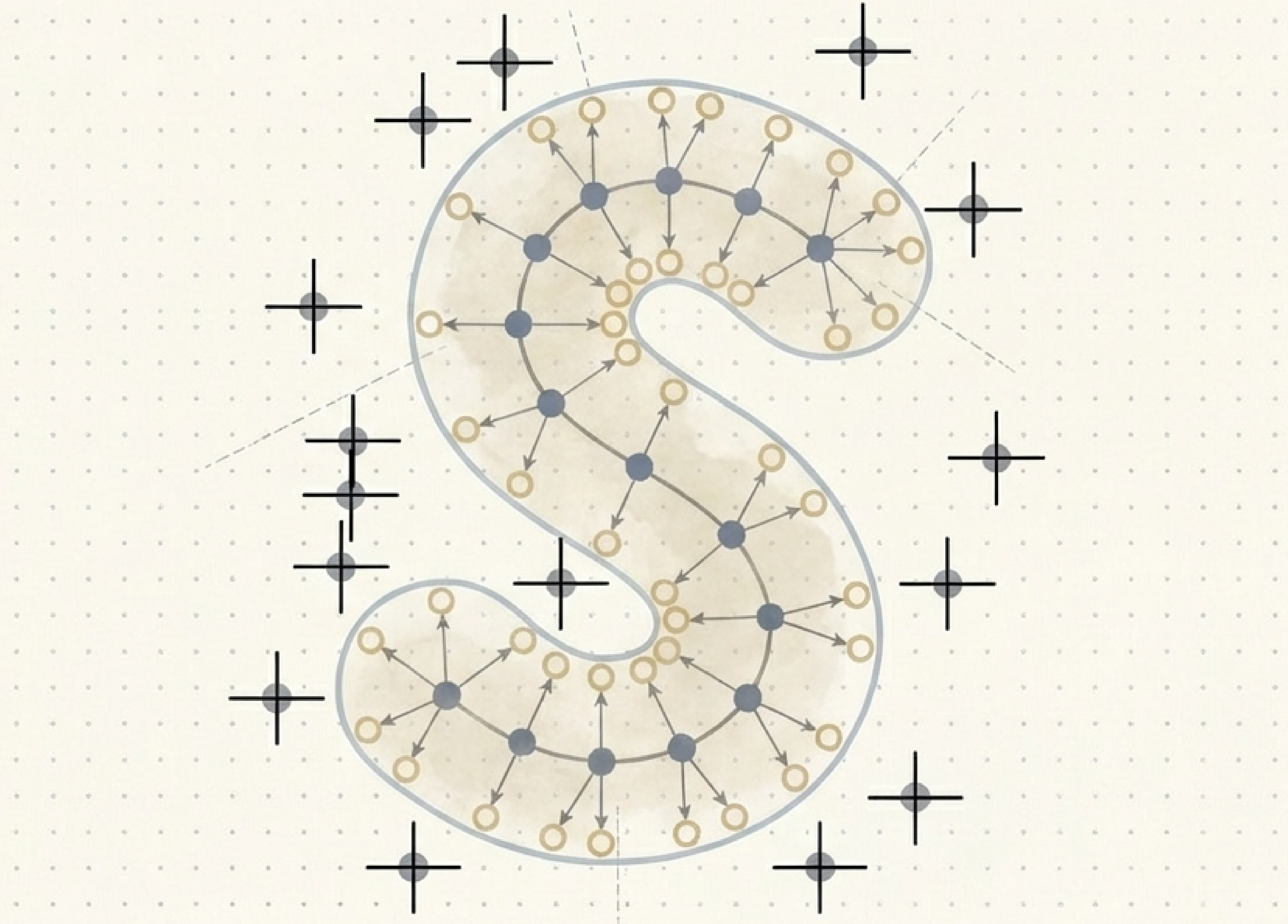
2. Border points are evaluated and subsequently assigned to the cluster of their nearest Core point, forming the protective skin of the shape.

Step 4: Leaving anomalous points in the void

No Forced Fits.

Unlike centroid algorithms that mathematically force every single outlier into the nearest cluster (drastically distorting the data), DBSCAN explicitly isolates anomalies.

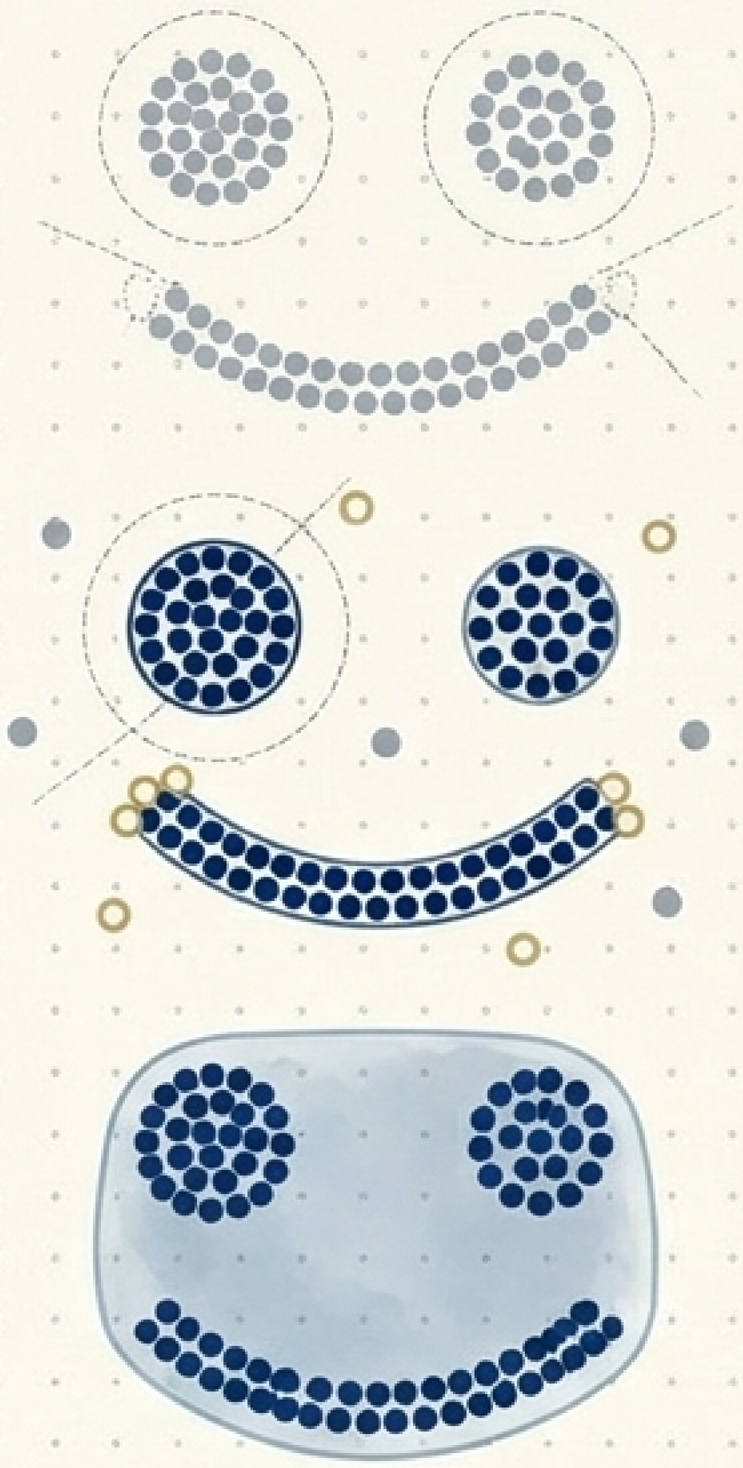
Noise points remain unclassified, preserving the integrity of the natural clusters.



Translating geometric theory into practical implementation

```
db = DBSCAN(eps=0.5, min_samples=5).fit(X)
labels = db.labels_
```

Note: Noise points are automatically assigned an output label of -1.



Strategic advantages and operational limits of DBSCAN

Strengths	Limitations
✅ **Anomaly Detection** Built-in robustness to outliers; natively identifies noise rather than distorting models.	⚠️ **Varying Density Flaw** Struggles if clusters have drastically different internal densities, as a single Epsilon value cannot suit both.
✅ **Shape Agnostic** Easily maps concentric circles, crescents, and complex non-spherical spatial data.	⚠️ **Hyperparameter Sensitivity** Minute, granular changes to the radius drastically alter macro-outputs.
✅ **Zero Pre-guessing** Automatically discovers the correct number of natural clusters without an elbow curve.	⚠️ **No Predict Function** Cannot cluster new, unseen data points without entirely retraining the model on the full dataset.

Clustering as an organic mapping of natural density

Density-Based

DBSCAN

Spatial Clustering of Applications with...

Noise

DBSCAN transforms clustering from a rigid search for statistical centres into an organic mapping of density, allowing the data's natural shape—and its anomalies—to reveal themselves.